

A continuous control paradigm for Motor Brain-Computer Interfaces

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Introduction: A large part of research on Motor Imagery Brain-Computer Interfaces (BCI) relies on variations of a cue-based paradigm. The user fixates on a target for a period, then is cued to perform or imagine the movement for a fixed duration, and then stops. This is a simple setup, but also results in very limited commands, making it impractical for Spinal Cord Injury or Stroke Patients.

We propose a new paradigm for the development of Motor BCIs. By simultaneously recording EMG and EEG of executed movements, we can continuously measure the ground truth movement of the user during a control task. This gives us a mapping of EEG to movement actions to be learned by Machine Learning methods. Previous research on BCI control treats control as a task without a ground truth, where performance is measured indirectly. In our setup, the exact control commands can be measured.

Methods: We record 32 channel EEG, 4 channel EOG, and 4 channel EMG data of 19 healthy participants. Each participant first performs a standard, cued calibration session moving their left and right hands. After this, participants perform a control task in which they steer a simulated car by moving their left and right hands. With this paradigm, we develop EEG decoding models based on the calibration session, and evaluate their quality with EMG as ground truth during the control task. This allows us to establish the exact movement onset and offset, forces models to distinguish between movement preparation and movement execution, forces models to be robust against EEG patterns arising from a more complex task environment, and forces models to be robust for both short and long movements.

We train a within-subject Common Spatial Patterns (CSP) classifier and a Convolutional Neural Network (CNN) using the cues from the calibration session, and evaluate whether they are able to generalise to the control task, classifying segments into left, right or rest. Additionally, we establish a regression task based on the EMG envelopes between the two hands, indicating overall steering strength for either direction at any time, which is tackled with a ridge regression model.

Results: Both classifiers perform well on the calibration session with an average F1 score (i.e. mean of precision and recall) of 0.7 for CSP and 0.63 for the CNN. The CNN failed to generalize to the control task, with an F1 score of 0.37, while the CSP performed reasonably well with an average F1 score of 0.53. With the CSP there were four subjects with an F1 score > 0.7 , suggesting this model is effective. The ridge regression model failed to meaningfully reconstruct the EMG envelope.

Conclusion: We introduce a new paradigm and dataset to support the development of robust control methods from Motor BCIs. This new dataset still has many avenues for exploration, such as decoding for short “bursts” of movement, as well as optimally estimating the strength, onset, duration and progression of the movement.

Highlights:

- We introduce a novel paradigm and dataset for Motor BCI control, where the exact control inputs of the user are available as ground truth for evaluation of EEG decoders.
- We treat this both as a classification (left/right/rest) and real-time regression (EMG envelope) task, and report results of models trained on a calibration session and how they perform during a control task.
- The dataset allows for much further analysis with movement bursts, estimating strength, exact movement onset, and overall progression of the movement, and is planned to be released in pairing with a journal publication in 2026.